Vouch

Peer Tutoring Marketplace with a Verifiable Reputation Record

CS 4800-02 · Fall 2026 · Option 04, Track A · Three-person team

Combined project document — Business Plan (ICE-1) and Project Backlog & Sprint Plan (ICE-2)

Working product name; substitute the team's chosen name throughout. All items marked TBD need a decision from the group.

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For the group: what needs your input

- **Names against the three roles** in Part I, Section 1.4. Ownership drives the whole sprint plan, so this is the first decision.
- **The product name.** Currently a placeholder.
- **The open question in Part II, Section 2** — one track or both. Worth asking the instructor in week 1, not week 4.
- **Whether the descoping decisions in Part II, Section 6 are ones we all agree to.** Cutting 318 of 531 hours is the central claim of this plan; if anyone disagrees with a specific cut, the trade has to come out of something else.

Part I — Business Plan

Part I describes the product as a business. Part II defines what this three-person team will actually build in twelve weeks. Where the two differ, Part II governs — the gap between them is the descoping record in Part II, Section 6, not an oversight.

1. Executive Summary

1.1 Business Concept

Vouch is a campus-scoped marketplace that connects students who need academic help with peer tutors who have already succeeded in the same courses. It differs from general tutoring platforms in two ways. First, matching is course-specific and availability-aware rather than subject-general — a student searching for help does not get a list of "math tutors," they get ranked peers who have completed the exact course, with the exact professor where possible, and whose free hours actually intersect their own.

Second, and more central to the product, the reputation record is tamper-evident. Reviews are written into an append-only log in which each entry cryptographically incorporates the hash of the entry before it. Altering or deleting a past review breaks the chain at that point and every verification afterward fails. Reviews can only be created against an attested session — a booking both parties confirmed took place — which closes the loop that lets fake reviews accumulate on open platforms.

The pitch to a campus is therefore not "we have tutors." It is "we have a reputation record you can audit."

1.2 Target Market

The initial market is a single large public commuter university, modeled on Cal Poly Pomona: roughly 30,000 students,

heavily represented in engineering, computer science, and business, with a high proportion of transfer and first-generation students and a documented set of high-DFW-rate gateway courses (calculus sequences, general chemistry, physics, data structures).

Expansion targets are institutions with the same shape — large CSU and regional public campuses where the official tutoring center is oversubscribed and the informal market already exists in Discord servers and group chats but is unstructured, unaccountable, and invisible to the institution.

The buyer and the user are not always the same. Students are the users. The paying customer in the institutional model is the department, learning center, or student success office that wants supplemental instruction capacity it can report on.

1.3 Proposed Solution

A web application with four subsystems:

1. **Profiles and availability.** Tutors declare completed coursework, grade earned, teaching modality, and a weekly availability model. Students declare current enrollment and learning preferences.
2. **Matching engine.** A deterministic, weighted scoring function ranks tutors against a structured help request. A language model handles only the intake step — turning a free-text request ("lost on dynamic programming, midterm Thursday") into structured fields — and the explanation step, generating the human-readable reason a given tutor was ranked where they were. It does not compute the ranking.
3. **Booking and attestation.** Requests become bookings; bookings become sessions; sessions require confirmation from both parties before a review can be written. Cancellation and no-show policy is defined in product terms; automatic enforcement is outside prototype scope (Part II, Section 6).
4. **Tamper-evident reputation ledger.** Reviews and attestations are appended to a hash-linked log. A verification routine walks the chain and reports the first broken link. The administrative interface includes a deliberate tamper action for demonstration and audit-testing purposes.

Post-session, generated summaries capture concepts covered and unresolved gaps, and those gaps feed back into the student's profile so subsequent matches improve. Summaries are advisory content and are stored outside the ledger; nothing that a model generates participates in hash computation or verification.

1.4 Team

The project is delivered by a three-person team across three four-week sprints. The project handout describes two sub-teams (Intelligence, and Reputation & Booking); three people cannot split into two meaningful sub-teams, so ownership is assigned by system area instead, with each member accountable for one vertical slice end to end.

Role	Member	Owns
Lead / Backend & Ledger	*TBD*	Booking lifecycle, attestation, hash chain, verification, disputes; sprint planning and demos
Data & Matching	*TBD*	Schema, course taxonomy, availability, matching score, LLM intake and summaries
Frontend & Quality	*TBD*	All interfaces, test plan, end-to-end tests, accessibility; documentation lead

Documentation deliverables are shared, with the Frontend & Quality role owning assembly and consistency. Capacity implications of the team size are worked through in Part II, Section 1.

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2. Product Overview

2.1 Core Features

The full feature set of the product. Which of these the team commits to, and in which sprint, is set out in Part II, Section 4.

Foundation

- Student and tutor profiles with verified course history
- Weekly availability model with timezone and recurring-block support

- Help request creation with structured fields
- Manual matching, so the workflow is exercised end-to-end before the engine exists

Engine and ledger

- Weighted matching engine with ranked results and explanations
- Booking lifecycle: request → accept → scheduled → completed
- Dual-party session attestation
- Review capture, anchored to an attested session
- Hash-chained append-only review log with a verification endpoint

Integrity and measurement

- Dispute filing, evidence attachment, and resolution records
- No-show policy with automatic penalty application
- Match quality metrics: acceptance rate, completion rate, repeat-booking rate, post-session confidence delta
- Administrative audit view with tamper demonstration

Matching score. Ranking is a transparent linear combination, tunable and defensible:

```
score = 0.40 · course_match
      + 0.30 · availability_overlap
      + 0.20 · tag_overlap
      + 0.10 · normalized_tutor_rating
      − 0.10 · current_load_penalty
```

`course_match` scores 1.0 for the same course code, 0.6 for the same subject at an adjacent level, and 0 otherwise. `tag_overlap` is the fraction of the request's topic tags the tutor declares. `availability_overlap` is the fraction of the student's requested window covered by the tutor's free blocks. The load penalty exists so that a small number of highly rated tutors do not absorb all demand and degrade response time.

The four positive weights sum to 1.0, so a score is directly interpretable as a percentage fit. Semantic embedding of topics and a learning-style term were both cut for capacity (Part II, Section 6); the taxonomy is designed so either can be added later without changing the score's shape.

2.2 Technology Stack

Layer	Choice	Rationale
Frontend	React + TypeScript, Tailwind	Component reuse, type safety across a small team
Backend	Node.js + Express (or FastAPI)	Fast iteration, broad team familiarity
Database	PostgreSQL	Relational integrity for bookings; append-only ledger table with a database-level rule preventing UPDATE and DELETE
Ledger	SHA-256 hash chaining in application code	Each record stores `prev_hash`; verification recomputes the chain
AI services	Hosted LLM API for structured intake, ranking explanations, and session summaries; embedding model for topic-to-taxonomy mapping	No training, no GPU, no labeled dataset required
Auth	Campus SSO in production; email + role-based access for the prototype	Institutional identity is what makes reviews meaningful
Testing	Jest / PyTest, plus a dedicated ledger-integrity suite	Verification correctness is the highest-value test target
CI / Hosting	GitHub Actions, containerized deploy	Sprint demo reliability

Model outputs shown in demonstrations are cached, so no live API call sits on the critical path of a graded presentation.

3. Market Analysis

3.1 Industry Overview

The global private tutoring market is large and growing, though published estimates vary widely by research firm — a range worth stating plainly rather than citing a single figure as fact. The Business Research Company places the market at approximately $131 billion in 2025, growing to roughly $143 billion in 2026 at about a 9.1% CAGR, and projects roughly $209 billion by 2030. IMARC Group estimates approximately $134 billion for 2025; Fortune Business Insights estimates roughly $67 billion for the same year using a narrower definition. The consistent signals across all of them are high single-digit to low double-digit growth, rapid migration from in-person to online delivery, and AI-driven personalization as the dominant product trend.

In the United States, roughly one in five students receives some form of private tutoring, and the large majority of tutoring programs now offer online delivery. Peer tutoring specifically sits in a gap: campus learning centers provide it free but with limited hours, limited course coverage, and long waits, while commercial platforms charge $40–$80 per hour, which is out of reach for the students who most need help.

3.2 Target Market & Competitive Analysis

Serviceable market at one campus. With approximately 30,000 students, assume 20% seek paid academic help in a given term (6,000 students), at an average of 6 sessions per term at $20 per session. That is roughly $720,000 in per-term transaction volume at a single institution. This is an order-of-magnitude estimate built from stated assumptions, not a forecast; each assumption should be tested against campus survey data before it is relied on.

Competitor	Model	Weakness Vouch addresses
Wyzant	National tutor marketplace, $40–$80/hr	Price; tutors have no course-specific or professor-specific context
Varsity Tutors	Subscription + on-demand	Not peer-based; no institutional integration
Knack	Campus peer tutoring, institution-funded	Closest competitor; reputation is platform-asserted rather than independently verifiable
Campus tutoring center	Free, staffed, limited hours	Capacity-constrained; narrow course coverage; no scheduling flexibility
Discord / GroupMe / word of mouth	Free, informal	No accountability, no record, no discovery beyond one's existing network

The realistic incumbent is the last row. Most peer tutoring already happens informally; the product does not need to create demand, it needs to make an existing behavior discoverable and accountable.

Defensibility. The hash-chained record is straightforward to replicate technically. The durable advantage is the accumulated reputation graph within an institution and, if achieved, integration with campus identity and course enrollment data — which is an institutional relationship rather than a piece of code.

4. Marketing and Sales Strategy

4.1 Marketing Channels

- **Course-level targeting.** Concentrate launch effort on the ten highest-DFW-rate courses rather than the campus as a whole. Density inside one course is worth more than thin coverage across two hundred.
- **Faculty and advisor referral.** Instructors of gateway courses are a credible, low-cost distribution channel and are motivated by their own pass rates.
- **Student organizations.** Discipline-specific clubs (SWE, ACM, NSBE, SHPE, business fraternities) already contain the supply side.
- **Existing informal channels.** Presence in the Discord servers and group chats where the behavior already occurs, positioned as infrastructure rather than as a replacement.

- **Physical presence.** Tabling during weeks 4–6 and 10–12, when midterm and final pressure peaks.

4.2 Customer Acquisition

Supply is seeded first. A marketplace with tutors and no students is recoverable; the reverse is not. Recruit 40–60 tutors in the top-ten course list before opening to students, using guaranteed-hours incentives and transcript-verified badges.

Acquisition follows a two-sided sequence:

1. **Supply seeding (weeks 1–4):** direct recruitment through faculty referral and club partnerships.
2. **Demand launch (weeks 5–8):** targeted at students in the same courses, timed to the first midterm.
3. **Referral loop:** both parties receive session credit for a referral that completes a first session.

Cost per acquired student in the campus model is near zero in cash and non-trivial in labor — the honest framing is that early acquisition is founder time, not marketing spend.

4.3 Growth Strategy & Monetization

Monetization options, in order of realism for a campus product:

1. **Transaction fee (primary).** A 12% platform fee on peer-to-peer sessions. At a $20 session that is $2.40 in revenue. At 400 sessions per month, monthly revenue is 400 × $2.40 = $960; at 2,000 sessions per month it is $4,800. Against roughly $200–$400 per month in hosting and API costs at that volume, the model reaches contribution-positive somewhere in the low hundreds of monthly sessions — the sensitivity is almost entirely to session volume, not to price.
2. **Institutional license (highest value).** An annual per-campus fee, in the $15,000–$40,000 range, sold to a student success office that wants supplemental instruction capacity plus reporting. One institutional contract is worth more than a full year of transaction fees at realistic student volume, which means the long-term business is B2B even though the user experience is B2C.
3. **Tutor subscription (secondary).** A monthly fee for profile promotion and analytics. Modest revenue; introduces a ranking-integrity conflict that should be handled explicitly — paid placement must be labeled and must not enter the match score.

Growth path. Establish depth in ten courses at one campus, expand course coverage to the full STEM core, then expand to a second and third campus in the same system where the course catalog and student profile are close enough that the taxonomy transfers with minimal rework. Multi-campus expansion is where the reputation ledger becomes a genuine asset: a portable, verifiable record that follows a tutor across institutions.

Key risks.

- *Cold start.* Mitigated by course-level concentration rather than campus-wide launch.
- *Disintermediation.* Once matched, both parties can transact off-platform. Mitigated by making the reputation record valuable enough that tutors want sessions logged, not by attempting to prevent contact.
- *Institutional policy.* Some campuses restrict paid tutoring by enrolled students in courses they have taken. This must be verified with the relevant office before any real launch and is a genuine unknown, not a formality.
- *Academic integrity.* Tutoring adjacent to coursework can shade into completing coursework. Policy, session summaries, and dispute records exist partly to make that boundary auditable.

Market figures in section 3.1 are drawn from published industry reports as of 2026 and should be re-verified before submission. Financial projections throughout are illustrative models built from stated assumptions rather than forecasts.

Part II — Project Backlog & Sprint Plan

1. Estimation Basis

Estimates are in **ideal engineer-hours** — uninterrupted work by one person who knows what they are doing. They are not calendar hours and do not include meetings, blocked time, or context switching.

Team capacity per sprint:

```
3 members × 8 hours/week × 4 weeks       =  96 raw hours
− 25% reserve (other coursework, exams,
  rework, integration friction, illness)  =  24 hours
──────────────────────────────────────────────────────
Committed capacity per sprint             =  72 hours
Committed capacity across 3 sprints       = 216 hours
Per person, per sprint                    =  24 hours
```

Total scope identified in the business plan: 531 hours. Against 216 hours of capacity, the full product is **146% over** what this team can build. Section 5 records what was removed.

The consequence worth stating plainly: a three-person team builds roughly 60% of what a five-person team builds, not 60% of the features. Fixed costs — environment setup, authentication, documentation, demo preparation — do not shrink with headcount, so they consume a much larger share of a smaller budget. Documentation alone is 39 of 216 hours here, versus 36 of 360 before.

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2. Team Structure and Ownership

Roles are defined in Part I, Section 1.4. In summary: Lead / Backend & Ledger, Data & Matching, and Frontend & Quality, with documentation shared.

Open item to resolve with the instructor before Sprint 1: whether a three-person group is expected to deliver both tracks of Option 04 or only one. If both tracks are required at the scope described in the handout, this plan is the honest maximum, and the instructor should see the capacity arithmetic in Section 1 rather than a plan that quietly promises more.

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3. Epics

Code	Epic
A	Infrastructure & Environment
B	Identity, Profiles & Course Data
C	Availability & Scheduling
D	Matching Intelligence
E	Booking & Session Attestation
F	Tamper-Evident Reputation Ledger
G	Disputes
H	Administration
I	Process & Documentation Deliverables
J	Quality, Testing & Non-Functional

Priority uses MoSCoW: **M** = Must (demo fails without it), **S** = Should.

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4. Sprint Plan

Sprint 1 — Weeks 1–4 · Foundation

Sprint goal: A running application where a tutor publishes availability, a student posts a structured help request, and an administrator matches them by hand. No intelligence, no ledger. The workflow is proven before either is built.

ID	Task	Owner	Est (h)	Pri
A1	Repository, branch strategy, PR template, CI (lint, test, build)	Lead	4	M
A2	Local dev environment, Docker Compose with PostgreSQL	Lead	4	M
A3	Application skeleton: React + TypeScript client, Express API	Frontend	5	M
A4	Single staging deployment and environment configuration	Lead	4	M
B1	Data model v1: users, courses, tutor skills, availability, requests	Data	7	M
B2	Authentication and role-based access (student / tutor / admin)	Lead	8	M
B3	Unified profile: create, read, update, with tutor coursework fields	Data	8	M
B4	Hand-seeded catalog of ten target courses, taxonomy v1	Data	2	M
C1	Availability model (recurring weekly blocks) and editor interface	Data	9	M
D0	Help request form with structured fields	Frontend	4	M
D1	Manual matching: admin assigns a tutor to a request	Frontend	3	M
I1	Software Requirements Specification	Shared	7	M
I2	Architecture document and component diagram	Shared	5	M
I3	Sprint 1 demo preparation and retrospective	Shared	3	M
	Sprint 1 total		**73**	

Sprint 2 — Weeks 5–8 · Engine and Ledger

Sprint goal: Matching becomes automatic and reviews become tamper-evident. This sprint contains the entire critical path and is the one that decides the grade.

ID	Task	Owner	Est (h)	Pri
E1	Booking lifecycle state machine: request → accept → scheduled → completed	Lead	10	M
E2	Booking interface: student requests, tutor accepts or declines	Frontend	8	M
E3	Dual-party session attestation	Lead	6	M
F1	Ledger schema with database-level append-only constraints	Lead	6	M
F2	SHA-256 hash chaining implementation	Lead	7	M
F3	Chain verification endpoint reporting the first broken link	Lead	7	M
F4	Review capture interface, anchored to an attested session	Frontend	6	M
D4	Weighted matching score implementation (deterministic)	Data	8	M
D5	Ranked results interface with visible score breakdown	Frontend	6	M
J1	Ledger integrity test suite	Data	6	M
I4	Sprint 2 demo preparation and retrospective	Shared	3	M
	Sprint 2 total		**73**	

Critical path: E1 → E3 → F4 → F1 → F2 → F3 → J1. Every item is required by the demo the handout names — book, complete, review, then attempt to alter a review and fail. None of it can be parallelized away, and with one person owning most of it, this chain has no redundancy. It starts in week 5, not week 7.

Single-owner risk. The Lead owns 36 of this sprint's 73 hours, all of it on the critical path. If that person loses a week, the sprint fails. Mitigation: the Data role pairs on F2 and F3 during week 6 so a second person can carry the hash chain if

needed. This is deliberate redundancy bought at the cost of a few hours, and it is the single most valuable trade in the plan.

Simplified matching. D4 scores on exact course-code match, tag overlap, availability intersection, and tutor rating. Semantic embeddings were cut (Section 5); the score stays a transparent weighted sum that can be defended on a slide.

Sprint 3 — Weeks 9–12 · Intelligence, Disputes, Delivery

Sprint goal: Add the language-model features, close the loop with disputes, and produce the graded artifacts. Deliberately the lightest sprint, because week 12 collides with finals.

ID	Task	Owner	Est (h)	Pri
D3	LLM structured intake: free text to validated fields, with schema validation	Data	8	M
D6	Generated ranking explanations, with response caching	Data	5	S
D7	Session summary generation, feeding gaps back into student profiles	Data	8	S
G1	Dispute filing and resolution, recorded to the ledger	Lead	9	M
H1	Administrative audit view with tamper demonstration control	Lead	8	M
J2	End-to-end test suite covering the full demo path	Frontend	6	M
J3	Accessibility and usability checklist pass	Frontend	4	S
I5	Test plan document	Shared	6	M
I6	Final report and user manual	Shared	7	M
I7	Final presentation and demo rehearsal	Shared	6	M
	Sprint 3 total		**67**	

Sprint 3 is planned 5 hours under capacity. That slack is intentional and should not be spent before week 11.

Sequencing note. The three language-model tasks (D3, D6, D7) are the only remaining AI work and all sit in the final sprint. This is deliberate — they are additive rather than structural, so if the team is behind, D6 and D7 can be dropped without breaking anything, and the project still demonstrates AI through D3. Putting them earlier would put the ledger at risk; putting the ledger later would risk the entire grade.

5. Capacity Summary

Sprint	Weeks	Committed (h)	Capacity (h)	Variance
Sprint 1	1–4	73	72	+1
Sprint 2	5–8	73	72	+1
Sprint 3	9–12	67	72	−5
Total	**1–12**	**213**	**216**	**−3**

Per person that is roughly 24 hours per sprint, or 6 hours of focused work per week each.

6. Descoping Decisions

Total identified scope was 531 hours; 213 fit. The 318 hours removed break into two groups: the 182 hours already deferred in the business plan (payments, campus SSO, native mobile, video conferencing, multi-campus tenancy, messaging, notifications, referrals, paid placement, learning-style instrument, calendar export), plus the 136 hours

below, cut specifically because the team is three people rather than five.

Category	Hours	What goes, and what replaces it
Matching sophistication	25	Semantic embeddings and taxonomy mapping, the learning-style term in the score, and the match-quality metrics dashboard. Replaced by exact course-code matching plus tag overlap, with raw counts shown in the admin view.
Booking depth	32	Rescheduling, double-booking and availability conflict detection, no-show policy with automatic penalties, and rate or pricing fields. The system supports one happy path; the no-show policy is documented rather than enforced in code.
Discovery & navigation	24	Tutor directory browsing, search and filtering, pagination and sorting across list views, and the in-app notification centre. Students reach tutors through the match flow only.
Review & dispute depth	19	Moderation queue, review edit window, evidence attachment on disputes, and rich-text session notes. Disputes are text records; the ledger integrity story is unaffected.
Profile & catalog scope	18	Separate student and tutor profile flows (unified into one), course catalog import (ten hand-seeded courses instead), and administrative override tooling beyond basic assignment.
Non-functional & environment	18	Dedicated security pass (folded into a pull-request review checklist), full accessibility audit (reduced to a 4-hour checklist), second deployment environment, and automated sprint metrics.
	136	

What was deliberately protected. Three things survived every round of cutting: the hash chain and its verification, the dual-party attestation that makes reviews meaningful, and the documentation deliverables. The first two are what the handout's demo tests. The third is graded directly and is the easiest thing to lose when a small team runs late.

7. Risks to the Plan

- **No bus factor.** With three people, every role is a single point of failure and the Lead's ownership of the critical path is the sharpest instance. The pairing on F2 and F3 in week 6 is the mitigation; it should not be the first thing cut when the sprint gets tight.
- **Sprints 1 and 2 have no slack.** Both are planned at +1 hour. There is no absorption capacity until Sprint 3, which means any slip propagates rather than dissipating. The first release valve in Sprint 1 is B4 (catalog seeding, 2h) and the second is D1 (manual matching, 3h).
- **The ledger is the demo.** If Sprint 2 is at risk, cut D5 polish and then D4 refinement before touching anything in E or F.
- **Documentation compresses under pressure.** I1, I2, I5, and I6 total 25 hours of graded deliverable, plus 12 hours of demo and presentation work. On a 216-hour budget that is 17% of everything the team will do, and it is the portion with no visible consequence for skipping until the grade arrives.
- **Sprint 3 collides with finals.** The 5-hour buffer exists for this and should be defended.